RSRefSeg 2: Decoupling Referring Remote Sensing Image Segmentation With Foundation Models

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***Abstract*—Referring remote sensing image segmentation (RRSIS) facilitates flexible scene analysis by leveraging vision-language collaborative interpretation. However, conventional coupled frameworks typically perform pixel decoding after cross-modal fusion, conflating target localization ("where") with boundary delineation ("how"). While recent decoupled approaches utilizing foundation models [e.g., segmentation foundation model (SAM)] attempt to separate these tasks, they predominantly rely on reductionist "box-to-mask" or "prompt-to-mask" pipelines. Such simple interfaces compress rich referring expressions into simplistic geometric constraints, severing semantic consistency and failing to leverage open-vocabulary visual-semantic alignment, particularly in multientity scenarios. To address these limitations, we propose RSRefSeg 2, a framework that logically reformulates the workflow into sequential subtasks of "slack localization" and "refined segmentation." Central to our approach is a novel cascaded second-order referring prompter, designed to construct a robust semantic bridge between CLIP's open-world understanding and SAM's segmentation capabilities. Specifically, we introduce an orthogonal subspace decomposition mechanism that separates text embeddings into complementary components. This enables implicit cascaded reasoning to isolate target attributes from background noise: first performing slack localization via cross-modal interaction to identify potential regions, and subsequently generating refined prompts to guide SAM's precise delineation. Furthermore, we incorporate parameter-efficient tuning to align natural image priors with remote sensing domains. Extensive experiments on RefSegRS, RRSIS-D, and RISBench demonstrate that RSRefSeg 2 significantly outperforms state-of-the-art methods, achieving an approximate 3% improvement in generalized intersection over union (gIoU) while offering superior diagnostic interpretability. The code is available at: https://github.com/KyanChen/RSRefSeg2**

***Index Terms*—Parameter-efficient fine-tuning (PEFT), referring image segmentation, remote sensing images, vision-language foundation models.**

I. Introduction

REFERRING remote sensing image segmentation (RRSIS) represents an advanced vision-language multimodal task that aims to achieve precise segmentation of semantic regions in remote sensing imagery guided by natural language descriptions [1], [2], [3]. This language-guided visual understanding mechanism addresses limitations of conventional semantic segmentation frameworks by offering enhanced flexibility and granularity in image analysis [4]. The paradigm demonstrates significant practical value in land-use planning, ecological monitoring, and disaster management applications [5], [6], [7], [8], [9], [10], [11].

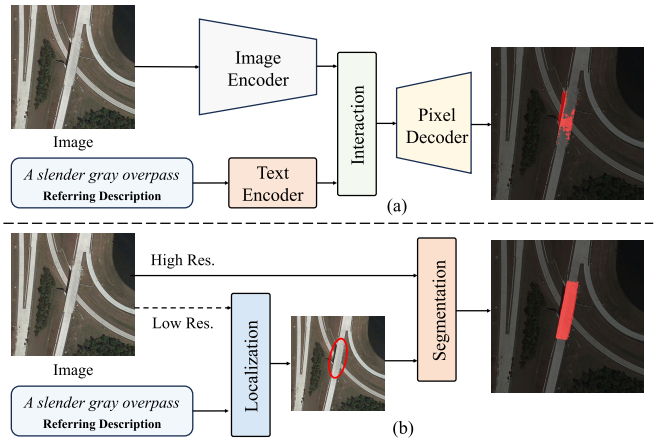


Fig. 1. (a) Existing coupled frameworks: end-to-end architecture that conflates localization and segmentation, limiting interpretability. (b) RSRefSeg 2: a decoupled framework adopting a sequential "localization-then-segmentation" pipeline.

Despite preliminary advances facilitated by benchmark datasets such as RefSegRS [1], existing methodologies face persistent limitations [12], [13]. Early CNN-RNN dual-encoder architectures achieved basic feature fusion but suffered from shallow interaction mechanisms that compromised pixel-level accuracy [1]. While attention mechanisms enhanced semantic interactions, mainstream approaches remain dependent on pretrained language models (e.g., BERT) and employ simplistic fusion strategies such as direct concatenation or unidirectional interactions [14], [15], [16]. These methods inadequately address distributional discrepancies between visual and textual features in remote sensing contexts. In addition, the scarcity of paired image–text datasets impedes fine-grained semantic alignment and limits cross-scene generalization [17], [18], [19], [20], [21], [22].

Existing frameworks predominantly employ a coupled architecture where pixel decoding after cross-modal fusion

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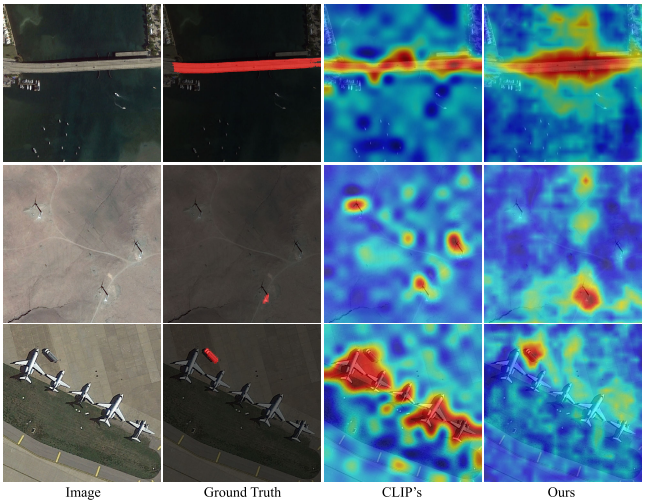


Fig. 2. Comparison of target perception between native CLIP and our proposed approach. While CLIP accurately activates target regions in simple scenarios, it fails when multiple objects of the same category exist (second row) or when primary scene entities differ from the referred target (third row), producing multiple or incorrect activation regions. Our RSRefSeg 2 utilizes a cascaded second-order prompting mechanism to decompose linguistic cues, effectively resolving ambiguity and precisely activating targets. Referring expressions: "the orange and gray large bridge in the middle," "the windmill at the bottom," and "the vehicle is on the right of the airplane on the upper left."

involving distinct encoders, interaction modules, and decoders [4], [23], [24], [25]. As illustrated in Fig. 1(a), this end-to-end coupling conflates the distinct subtasks of localization and segmentation, thereby limiting model performance and interpretability [24], [26]. Recently, the integration of foundation models has catalyzed a paradigm shift toward decoupled architectures [see Fig. 1(b)], exemplified by methods combining multimodal detectors or prompt generators with segmentation foundation model (SAM) [24], [27], [28], [29], [30]. While these approaches successfully disentangle localization from segmentation, they typically rely on a reductionist "box-to-mask" or "prompt-to-mask" pipeline. In this schema, complex referring expressions are compressed into preliminary geometric bounding boxes or simplistic semantic prompts to trigger SAM. They encounter intrinsic bottlenecks: 1) semantic information loss: compressing referring expressions, which are rich in attributes and spatial relationships, into simple geometric coordinates discards critical contextual nuances required for fine-grained disambiguation; and 2) weak cross-modal alignment: naïve prompt generation networks trained from scratch often suffer from a deficiency in open-vocabulary reasoning, thereby failing to effectively leverage the open-world visual-semantic alignment inherent in models like CLIP. Consequently, despite being architecturally decoupled, these methods fail to bridge the semantic gap between high-level linguistic comprehension and pixel-level delineation.

To bridge this gap, this article introduces *RSRefSeg 2*, a framework that advances beyond standard two-stage architectures (coarse-grained localization followed by fine-grained mask generation) by incorporating a cascaded second-order referring prompter. Unlike "box-to-mask" approaches that rely on shallow detection results, or existing "prompt-to-mask" methods that neglect open-set visual-semantic alignment, our framework constructs a deep semantic bridge between CLIP's open-world understanding and SAM's segmentation capabilities. Specifically, we propose a novel prompting mechanism that decomposes text embeddings into complementary semantic subspaces under orthogonal constraints. This design enables implicit cascaded reasoning: the first performs "slack localization" via cross-modal interaction to identify potential regions, while the second utilizes activated visual features to generate "refined segmentation" prompts. This two-step process effectively mitigates CLIP's tendency for misactivation in multientity scenarios (see Fig. 2). By transforming localization from a purely geometric task into a semantic alignment process, RSRefSeg 2 preserves fine-grained textual logic before guiding SAM. Furthermore, we integrate low-rank adaptation (LoRA) to efficiently address the domain shift from natural images to remote sensing imagery, ensuring robust adaptation while retaining generalizable knowledge.

The primary contributions are threefold as follows.

1) We propose RSRefSeg 2, a dual-stage decoupled framework that enhances interpretability by decomposing referring segmentation into sequential subtasks of "slack localization" and "refined segmentation," effectively leveraging cross-modal priors from vision foundation models.
2) We design a cascaded second-order prompting mechanism that resolves CLIP's semantic ambiguity in multientity contexts. By decomposing text embeddings into orthogonal subspaces for implicit cascaded reasoning, we establish a robust semantic bridge between CLIP and SAM, significantly outperforming conventional multimodal fusion strategies.
3) Extensive experiments across three benchmarks (RefSegRS, RRSIS-D, and RISBench) demonstrate that our method achieves superior performance in both segmentation accuracy and semantic comprehension, with ablation studies confirming the architectural coherence of the proposed components.

II. Related Works

A. Referring RRSIS

RRSIS represents a significant advancement in visual-language multimodal understanding, enabling precise segmentation of semantically relevant regions in remote sensing imagery through natural language descriptions [2], [19], [23]. However, two primary challenges persist: 1) inherent characteristics of remote sensing images, including significant scale variations, complex scene compositions, and low foreground–background contrast, complicate precise object boundary delineation [31], [32], [33], [34], [35]; and 2) accurate cross-modal alignment is required to interpret fine-grained semantic relationships between geospatial contexts and textual descriptions [36], [37], [38], [39].

Early approaches predominantly employed dual-stream architectures combining CNNs and RNNs for pixel-level predictions through shallow feature fusion [40]. Subsequently, a three-stage "representation-fusion-segmentation"

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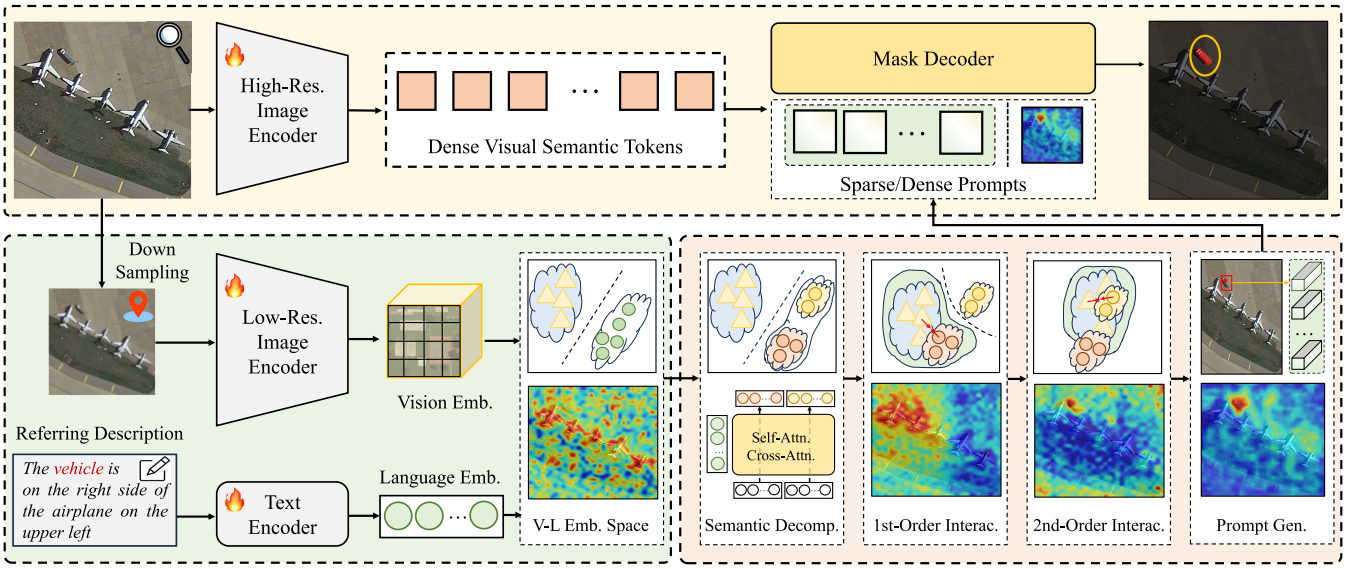


Fig. 3. RSRefSeg 2 comprises three principal components: the dual-modality aligned semantic feature encoder ( ), the cascaded second-order referring prompter ( ), and the prompt-guided refinement mask generator ( ).

paradigm was established, where modality-specific features from pretrained language models and visual encoders undergo multimodal fusion to generate segmentation masks [5], [6], [41]. Current research on cross-modal interaction mechanisms can be categorized into three approaches: 1) simple concatenation methods directly align textual and visual features, exemplified by LAVT's integration of BERT-derived features through visual transformers [40] and LGCE's cross-layer adaptive fusion [1]; 2) text-guided visual representation approaches modulate visual features using referring information, such as RMSIN's multiscale interaction modules [5] and FIANet's semantic decomposition networks [6]; and 3) bidirectional frameworks refine language representations through visual feedback, with architectures like BTDNet [12], CroBIM [3], and SBANet [25] employing bidirectional attention and scale-aware queries.

Unlike natural imagery where models such as CLIP achieve robust vision-language alignment, the substantial modality gap in remote sensing constrains existing methods [42], [43], [44]. Limited training data further impedes fine-grained semantic alignment, resulting in suboptimal feature generalization across diverse geospatial contexts [45], [46]. To address these limitations, RSRefSeg 2 introduces a decoupled "localization-segmentation" architecture leveraging prealigned vision-language models for knowledge transfer, thereby enhancing performance while enhancing interpretability compared to conventional frameworks.

B. Foundation Models

The rapid advancement of foundation models has catalyzed a paradigm shift in artificial intelligence, characterized by exceptional task-processing capabilities and cross-domain generalization properties [47], [48], [49], [50]. While text-driven large language models dominate current foundation model systems, vision foundation models demonstrate significant technological progress [17], [51]. CLIP achieves powerful zero-shot recognition through large-scale image–text contrastive learning [17], while Diffusion models surpass traditional GANs in image generation quality via progressive denoising architectures [32], [52]. SAM addresses universal image segmentation challenges through automated annotation engines [51], [53]. Remote sensing foundation models have undergone accelerated development [54], [55], [56], [57], [58], with the RingMo series achieving superior multitask performance through billion-parameter designs [59], [60], SpectralGPT and HyperSigma extending capabilities to multispectral and hyperspectral data [61], [62], and DynamicVis balancing precision and efficiency through token-reduction state-space architectures [18].

Pretrained model fine-tuning reduces sample dependency in high-precision scenarios while enhancing cross-scenario generalization [54], [63]. Representative applications include AnyChange for arbitrary change detection [64], CrossEarth as the first domain-generalized semantic segmentation model in remote sensing [65], and SegEarth-R1, which achieves semantic-guided segmentation through hierarchical visual-language encoder collaboration [38].

The proposed RSRefSeg 2 framework advances foundation model fine-tuning by integrating knowledge from general vision models to enhance referring segmentation performance under open-vocabulary text descriptions and open-set remote sensing scenarios. Task-specific optimization is achieved through knowledge transfer mechanisms under constraints of limited data and computational resources, providing practical solutions to real-world challenges, including sample scarcity and computational bottlenecks.

C. Parameter-Efficient Fine-Tuning (PEFT)

Large-scale foundation models have surpassed traditional machine learning paradigms that predominantly relied on fully supervised learning. Foundation models, including vision-language models (CLIP [17] and BLIP [66]) and vision

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models (DINO [67] and SAM [51]), demonstrate exceptional transferability and generalization capabilities across diverse downstream tasks. Within this context, prompt learning has emerged as a parameter-efficient adaptation strategy that bridges semantic discrepancies between pretrained models and task-specific requirements through strategic input design while preserving base model parameters [49], [68]. Derived from natural language processing, prompt learning elicits latent knowledge representations through designed input cues, manifesting in visual domains through *text prompt optimization* for vision-language models and *visual prompt optimization* for pure vision architectures [69], [70], [71], [72].

PEFT methodologies have gained prominence through architectural innovations such as adapters, LoRA, and prefix tuning, enhancing few-shot adaptation by integrating lightweight modules while maintaining frozen backbone parameters [73], [74]. CoOp and VPT demonstrate significant improvements in few-shot classification [68], [75], while RSPrompter equips SAM with automated instance segmentation capabilities for remote sensing imagery through prompt-guided adaptation [54].

A decoupled architecture for referring RRSIS with foundation models is introduced in this work. Cross-domain knowledge transfer is achieved through collaborative adaptation of CLIP and SAM architectures, where CLIP processes semantic referring information while SAM generates segmentation masks. A novel cascaded second-order prompting mechanism propagates contextual information between localization and segmentation phases, enabling joint optimization of both foundation models. This approach significantly reduces trainable parameters compared to full fine-tuning while improving segmentation accuracy in complex scenarios characterized by occlusions, ambiguous boundaries, and extreme scale variations.

D. Foundation Models for RRSIS

Most RRSIS approaches traditionally perform multimodal interaction within a shared latent space, followed by direct pixel-wise decoding, as shown in Section II-A. However, the paradigm has recently shifted toward leveraging pretrained foundation models to address domain gaps and cross-modal alignment. CLIP-driven methods, such as CD2FSAN [76] and CRIS [77], attempt to bridge the modality gap by aligning hierarchical visual features with linguistic descriptions. Despite their effectiveness in feature adaptation, these CLIP-based approaches often fail to disentangle target attributes (e.g., color and shape) from background noise and lack the granularity required for precise pixel-level boundary delineation, as they prioritize category-level alignment.

To improve segmentation quality, recent research has explored SAM-centered and generative architectures. SAM-based methods, such as Enti-TwistNet and RS2-SAM2 [24], [27], [28], [29], [30], focus on generating effective prompts from linguistic inputs to leverage SAM's zero-shot capabilities. However, these pipelines face significant challenges: *semantic compression*, where complex expressions are degraded into simplistic geometric prompts, and *weak grounding* due to scratch-trained generators. While generative models like diffusion and Seq2Seq architectures [4], [13], [38] capture fine-grained semantics, they suffer from a severe efficiency-accuracy tradeoff. Furthermore, attempts to collaborate across multiple foundation models often rely on rigid interfaces without deep semantic propagation, resulting in disjointed localization and segmentation.

Building upon our preliminary conference version, RSRefSeg [2], which combined CLIP's alignment with SAM via a naïve "feature-to-prompt" conversion, we address the lack of reasoning capability for ambiguous expressions inherent in previous designs. To overcome the limitations of rigid model integration, this work introduces a cascaded second-order prompter. By decomposing text embeddings into orthogonal subspaces, we effectively separate "slack localization" from "refined segmentation." This approach mitigates CLIP's misactivation in multientity scenarios and ensures robust semantic propagation across the foundation model pipeline.

III. Methodology

A. Overview

This article introduces RSRefSeg 2, a framework that decomposes the complex segmentation process into two sequential stages: coarse-grained object localization and fine-grained mask generation. By leveraging CLIP's cross-modal alignment capabilities and SAM's segmentation strengths, the framework enhances semantic understanding for open-vocabulary referring expression segmentation in remote sensing imagery. As illustrated in Fig. 3, the architecture comprises three principal components: a dual-modality aligned semantic feature encoder, a cascaded second-order referring prompter, and a prompt-guided refinement mask generator.

The dual-modality encoder processes textual referring expressions and visual features from low-resolution images within a prealigned semantic space, establishing unified cross-modal representations. The cascaded second-order referring prompter generates spatially activated localization information through semantic deconstruction of text embeddings and implicit multimodal feature interactions, which are encoded as spatially aware prompts. Finally, the refinement mask generator synthesizes pixel-level segmentation masks by integrating spatial details from high-resolution images with prompt-derived contextual information. The processing pipeline is formally defined as

$$\begin{aligned} \mathcal{F} &= \Phi_{\text{enc}}(\mathcal{I}_1, \mathcal{T}) \\ \mathcal{P} &= \Phi_{\text{prompter}}(\mathcal{F}) \\ \hat{\mathcal{M}} &= \Phi_{\text{dec}}(\mathcal{I}_2 \mathcal{P}) \end{aligned} \tag{1}$$

where $\mathcal{I}_1 \in \mathbb{R}^{H_1 \times W_1 \times 3}$ and $\mathcal{I}_2 \in \mathbb{R}^{H_2 \times W_2 \times 3}$ denote the downsampled low-resolution and original high-resolution images, respectively, while $\mathcal{T} \in \mathbb{R}^N$ represents the textual referring description. The dual-modality encoder $\Phi_{\text{enc}}(\cdot)$ encodes image and text features to produce $\mathcal{F} = \{v, t\}$, where $v \in \mathbb{R}^{h_1 \times w_1 \times d_1}$ corresponds to the visual feature map and $t \in \mathbb{R}^{(L+1) \times d_1}$ encapsulates textual features comprising L semantic tokens and one global <EOS> token.

Hybrid prompts $\mathcal{P}$ are generated by $\Phi_{\text{prompter}}(\cdot)$ through integration of the prealigned visual-text features, combining sparse

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positional embeddings (point/box coordinates) with dense segmentation probability maps. These prompts are utilized by the decoder $\Phi_{\text{dec}}(\cdot)$ to generate refined binary masks $\hat{\mathcal{M}} \in \mathbb{R}^{H_2 \times W_2}$ through fusion of high-resolution spatial information from $\mathcal{I}_2$ with prompt-guided contextual cues.

This hierarchical architecture reduces cross-modal alignment complexity through feature decoupling while preserving spatial details via high-resolution processing. The framework exploits SAM's prompt adaptability to maintain segmentation accuracy under computational constraints ($H_1 < H_2$, $W_1 < W_2$). By decoupling semantic interpretation from mask refinement, the design effectively balances open-vocabulary textual understanding with precise pixel-level segmentation.

B. Dual-Modality Aligned Semantic Feature Encoder

RSRefSeg 2 leverages the text–visual joint semantic space of the pretrained CLIP to enhance cross-scene alignment capabilities for open-set textual referring by extracting prealigned textual and visual features. While CLIP provides robust encoding of free-form text and visual inputs with established cross-modal semantic alignment [17], two critical challenges limit its performance in referring RRSIS. First, the complex characteristics of remote sensing images, including intricate spatial relationships and attribute descriptions within referring texts, significantly deviate from natural image description paradigms [78]. Second, coarse-grained image-sentence alignment often results in imprecise target localization, manifesting as misalignment across multiple targets or incomplete activation [79]. To address these limitations, an LoRA method is employed to optimize cross-modal feature alignment

$$\begin{aligned} t &= \Phi_{\text{enc-t}}\left(\mathcal{T}, W_t, W_t^{\star}\right) \\ v &= \Phi_{\text{enc-v}}\left(\mathcal{I}_1, W_v, W_v^{\star}\right) \\ W^{\star} &= AB^T \end{aligned} \tag{2}$$

where $\Phi_{\text{enc-t}}$ and $\Phi_{\text{enc-v}}$ denote CLIP's text and visual encoders, respectively. $W \in \mathbb{R}^{d \times d}$ represents the frozen original parameter matrix, while $W^{\star}$ corresponds to trainable low-rank parameters composed of $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{d \times r}$, with $r \ll d$ enforcing the rank constraint [80]. Through this decoupled design, RSRefSeg 2 shifts focus from enhancing CLIP's fine-grained alignment capabilities for remote sensing to address target slack localization, with final segmentation accuracy then achieved through a dedicated downstream module optimized for fine-grained region delineation.

To meet the dense feature requirements for referring segmentation, the representational capacity of CLIP is enhanced beyond its standard sparse image and text representations. The visual encoder is modified by removing the pooling layer to preserve spatial resolution, generating a dense visual feature map $v \in \mathbb{R}^{h_1 \times w_1 \times d_1}$. For textual processing, word-level features $t_{\text{word}} \in \mathbb{R}^{L \times d_1}$ are extracted by aggregating semantic embeddings from each token in the final hidden states, explicitly encoding fine-grained details including category, attribute, and spatial cues from the referring expression. The sentence-level embedding $t_{\text{sent}} \in \mathbb{R}^{1 \times d_1}$ is extracted from the end-of-sequence (EOS) token position to capture global textual semantics. The final text representation $t = [t_{\text{word}}, t_{\text{sent}}]$ is obtained through concatenation, achieving cohesive integration of local and global textual semantics with spatially preserved visual features for dense prediction tasks.

C. Cascaded Second-Order Referring Prompter

Unlike conventional two-stage methods that regress explicit bounding box coordinates or masks as prompts, the RSRefSeg 2 prompter is designed to generate semantic-rich dense prompts and embedding-based sparse prompts. This design aims to preserve high-dimensional semantic cues, such as location and shape priors, as well as response strengths to specific attributes, which are crucial for guiding SAM in complex, low-contrast remote sensing scenes. The RSRefSeg 2 prompter activates visual features relevant to textual descriptions within a prealigned text–visual semantic space to generate coarse localization prompts.

The design of the cascaded second-order referring prompter is motivated by the inherent limitations of CLIP in processing complex RRSIS queries. While native CLIP excels at global image-sentence alignment, it tends to prioritize core semantic entities (e.g., "airplane") while frequently overlooking the fine-grained constraints imposed by attribute descriptors (e.g., "white") and relative spatial relationships (e.g., "to the right of"). This bias is particularly pronounced in multientity remote sensing imagery, where dominant categorical signals often cause the visual encoder to indiscriminately activate all instances of the same category (misactivation). This phenomenon persists even in queries involving multicategory indications, as the weaker constraint signals necessary for unique identification are typically overshadowed by class-agnostic features [81], [82].

To mitigate these issues, we propose a "decompose-and-refine" methodology rooted in subspace learning [83], [84], [85]. We hypothesize that a referring expression can be decoupled into distinct yet complementary semantic subspaces. Specifically, we focus on a dual-subspace structure enforced via orthogonality constraints [see (11)] to ensure semantic separation [86], [87]. Theoretically, the first subspace is designed to capture dominant categorical information (facilitating broad activation), while the second isolates residual constraint information (enabling specific entity identification). Building on this decomposition, we employ a cascaded reasoning mechanism. We argue that a naïve parallel interaction strategy fails to effectively suppress background noise, as constraint signals are often spatially ambiguous without categorical anchoring. Consequently, our cascaded design mimics the human cognitive process of "locate-then-filter" [88], [89]. The first order performs "coarse scoping" to activate potential entities, while the second functions as "iterative visual filtering," where constraint features interact specifically with the activated regions from the first stage to implicitly prune nontarget instances. Since explicit labels for intermediate reasoning steps are unavailable, this mechanism, termed *implicit cascaded reasoning*, is achieved through the joint optimization of the subspace orthogonality constraints and the ultimate segmentation objective. This design also establishes an extensible interface for potential reinforcement learning-based optimization. In detail, the prompter comprises four key modules: Referring semantic

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decomposition, first-order text–visual interaction, second-order text–visual interaction, and sparse-dense prompt generation

$$\begin{aligned} t_1, t_2 &= \Phi_{\text{decomp}}\left(t_{\text{word}}\right) \\ t_1', v_1' &= \Phi_{\text{interact}}\left(t_1, v\right) \\ t_2', v_2' &= \Phi_{\text{interact}}\left(t_2, v_1'\right) \\ \mathcal{P} &= \Phi_{\text{prompt}}\left(t_2', v_2', t_{\text{sent}}\right) \end{aligned} \tag{3}$$

where $t_{\text{word}} \in \mathbb{R}^{L\times d_1}$ represents word-level text embeddings, $v \in \mathbb{R}^{h_1\times w_1\times d_1}$ denotes input visual features, and $t_{\text{sent}} \in \mathbb{R}^{1\times d_1}$ corresponds to sentence-level text embeddings. The complementary semantic subspace embeddings are denoted by $t_1, t_2 \in \mathbb{R}^{n_t\times d_1}$, with t_1', v_1' and t_2', v_2' representing text embeddings and visual features activated by first-order and second-order interactions, respectively. The generated prompts $\mathcal{P}$ comprises sparse prompt $p_{\text{sparse}} \in \mathbb{R}^{n_p\times d_2}$ and dense prompt $p_{\text{dense}} \in \mathbb{R}^{(H_2)/4\times(W_2)/4}$. Φ_{decomp}, Φ_{interact}, and Φ_{prompt} denote the referring semantic decomposition, text–visual interaction activation, and prompt generation modules, respectively.

1) Referring Semantic Decomposition: The decomposition process Φ_{decomp} decouples the original referring expression into two distinct semantic subspaces via cross-attention between predefined learnable vectors and the original referring embedding. To enhance joint optimization and feature representation learning, a bidirectional attention interaction module facilitates information exchange between subspace embeddings

$$\begin{aligned} t_1^q &= \Phi_{\text{ln}}\left(\Phi_{\text{self-attn}}\left(t_1^q\right)\right) \\ t_1 &= \Phi_{\text{ln}}\left(\Phi_{\text{mlp}}\left(\Phi_{\text{ln}}\left(\Phi_{\text{cross-attn}}\left(t_1^q, t_{\text{word}}\right)\right)\right)\right) \\ t_1 &= \Phi_{\text{ln}}\left(\Phi_{\text{mlp}}\left(\Phi_{\text{ln}}\left(\Phi_{\text{cross-attn}}\left(t_1, t_2\right)\right)\right)\right) \end{aligned} \tag{4}$$

where $t_1^q \in \mathbb{R}^{n_t\times d_1}$ denotes a preset learnable subspace text embedding, and the second subspace embedding t_2 follows an identical structure. Here, $\Phi_{\text{self-attn}}$, $\Phi_{\text{cross-attn}}$, Φ_{ln}, and Φ_{mlp} represent self-attention, cross-attention, layer normalization, and projection operations, respectively. N_{decomp} identical blocks are stacked to enhance decomposition capability.

2) Text–Visual Interaction: The decomposed semantic subspaces undergo cascaded interaction to enhance visual information expression relevant to referring descriptions. An attention feature activation module Φ_{interact} utilizes visual features as queries and semantic embeddings as keys and values

$$\begin{aligned} t_1' &= \Phi_{\text{ln}}\left(\Phi_{\text{self-attn}}\left(t_1\right)\right) \\ v_1' &= \Phi_{\text{ln}}\left(\Phi_{\text{mlp}}\left(\Phi_{\text{ln}}\left(\Phi_{\text{cross-attn}}\left(v, t_1'\right)\right)\right)\right) \end{aligned} \tag{5}$$

where the second subspace undergoes analogous processing with v_1' as input queries, establishing a cascaded structure. By stacking N_{interact} layers, a second-order text–visual semantic interaction framework yields refined visual features for precise target localization.

3) Sparse-Dense Prompt Generation: SAM requires prompts for segmentation, which are categorized as sparse prompts (e.g., points and bounding boxes) or dense prompts (e.g., coarse masks) [51]. However, SAM lacks inherent semantic understanding and cannot automatically generate segmentations based solely on referring text [54]. To address this limitation, we propose generating both sparse and dense prompts (Φ_{prompt}) from cross-modal feature interactions to effectively bridge textual descriptions with visual segmentation targets.

For sparse prompts, embedding representations are generated instead of direct point coordinates to enhance computational efficiency. The sparse prompt $p_{\text{sparse}} \in \mathbb{R}^{n_p\times d_2}$ is derived from cascaded-interacted visual feature v_2' and text feature t_2' through the following process:

$$\begin{aligned} p_{\text{sparse}}^q &= \Phi_{\text{ln}}\left(\Phi_{\text{self-attn}}\left(p_{\text{sparse}}^q\right)\right) \\ p_{\text{sparse}} &= \Phi_{\text{ln}}\left(\Phi_{\text{mlp}}\left(\Phi_{\text{ln}}\left(\Phi_{\text{cross-attn}}\left(p_{\text{sparse}}^q, t_2'\right)\right)\right)\right) \\ p_{\text{sparse}} &= \Phi_{\text{ln}}\left(\Phi_{\text{mlp}}\left(\Phi_{\text{ln}}\left(\Phi_{\text{cross-attn}}\left(p_{\text{sparse}}, v_2'\right)\right)\right)\right) \end{aligned} \tag{6}$$

where a randomly initialized query vector p_{sparse}^q sequentially undergoes self-attention, followed by cross-attention with text feature t_2' and visual feature v_2'. This process is iterated $N_{\text{p-gen}}$ times to progressively refine the prompt semantics before transforming them into sparse positional encoding prompts compatible with SAM.

The dense prompt $p_{\text{dense}} \in \mathbb{R}^{(H_2)/4\times(W_2)/4}$ is generated as an activation probability map through multimodal interaction between visual feature v_2' and whole-sentence text semantic t_{sent} as follows:

$$\begin{aligned} F_{\text{dense}} &= \Phi_{\text{norm}}\left(\Phi_{\text{up}}\left(\Phi_{\text{conv}}\left(v_2'\right)\right)\right) \\ t_{\text{sent}} &= \Phi_{\text{norm}}\left(t_{\text{sent}}\right) \\ p_{\text{dense}} &= \Phi_{\text{einsum}}\left(F_{\text{dense}}, t_{\text{sent}}\right) \end{aligned} \tag{7}$$

where Φ_{conv} applies a 3×3 convolutional kernel with BatchNorm-ReLU layers for feature smoothing, while Φ_{up} upsamples the low-resolution CLIP features ($h_1 \times w_1$) to SAM-compatible dimensions ($(H_2)/4 \times (W_2)/4$) via bilinear interpolation. Φ_{norm} performs L2-normalization, and Φ_{einsum} computes the dot product between the processed feature map $F_{\text{dense}} \in \mathbb{R}^{(H_2)/4\times(W_2)/4\times d_1}$ and text embedding $t_{\text{sent}} \in \mathbb{R}^{1\times d_1}$ to generate the activation probability map p_{dense}. The resulting sparse and dense prompts are jointly fed into SAM for mask generation.

D. Prompt-Guided Refinement Mask Generator

Although SAM demonstrates exceptional generalization capabilities for segmentation tasks, it lacks inherent task-specific perceptual awareness. Rather than directly augmenting SAM's semantic understanding, RSRefSeg 2 adopts a decoupled paradigm that leverages SAM exclusively as a prompt-based mask generator to fully exploit its general object segmentation capabilities. Empirical analysis reveals that completely freezing SAM's parameters yields suboptimal performance, primarily due to the substantial domain discrepancy between natural images and remote sensing data. To address this domain shift while minimizing training overhead and preserving SAM's generalizable segmentation knowledge, low-rank trainable parameters are incorporated into the ViT backbone, and only the lightweight mask decoder is fine-tuned. This adaptation process is formalized as

$$\begin{aligned} F_{\text{img}} &= \Phi_{\text{img-enc}}\left(\mathcal{I}_2, W_{\text{img}}^{\star}\right) \\ F_{\text{dense}} &= \Phi_{\text{prompt-enc}}\left(p_{\text{dense}}\right) \end{aligned}$$

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$$F_{\text{out}} = \Phi_{\text{concat}}\left(T_{\text{filter}}, T_{\text{IoU}}, p_{\text{sparse}}\right)$$
$$\hat{\mathcal{M}} = \Phi_{\text{mask-dec}}\left(F_{\text{img}} + F_{\text{dense}}, F_{\text{out}}\right) \tag{8}$$

where $\Phi_{\text{img-enc}}$, $\Phi_{\text{prompt-enc}}$, and $\Phi_{\text{mask-dec}}$ represent SAM's image encoder, prompt encoder, and mask decoder, respectively. $W^{\star}_{\text{img}}$ denotes the learnable low-rank parameters integrated into the image encoder according to (2). $\mathcal{I}_2 \in \mathbb{R}^{H_2\times W_2\times 3}$ is the input high-resolution image, while $F_{\text{img}} \in \mathbb{R}^{h_2\times w_2\times d_2}$ represents the resulting feature map. The sparse prompt $p_{\text{sparse}} \in \mathbb{R}^{n_p\times d_2}$ and coarse mask prompt $p_{\text{dense}} \in \mathbb{R}^{(H_2)/4\times(W_2)/4}$ are derived from the preprocessing flow, with $F_{\text{dense}} \in \mathbb{R}^{h_2\times w_2\times d_2}$ corresponding to the prompt encoder output. $T_{\text{filter}} \in \mathbb{R}^{4\times d_2}$ and $T_{\text{IoU}} \in \mathbb{R}^{1\times d_2}$ are predefined learnable tokens for mask filtering and IoU prediction, concatenated via Φ_{concat} to produce the multimask output $\hat{\mathcal{M}} \in \mathbb{R}^{4\times H_2\times W_2}$. The first mask in $\hat{\mathcal{M}}$ is selected as the final prediction. Additional technical details are available in the original SAM paper [51].

E. Loss Function

The overall loss function is formulated as the combination of three complementary components: $\mathcal{L}_{\text{seg}}$, a supervision loss for referring segmentation that integrates binary cross-entropy (CE) and Dice loss applied to the final predicted mask [2], [90]; $\mathcal{L}_{\text{ortho}}$, an orthogonal constraint loss that enforces semantic separation between two subspaces to maximize their semantic distance; and $\mathcal{L}_{\text{align}}$, an alignment loss that promotes fine-grained alignment between first-stage visual features and semantic features to facilitate knowledge transfer from CLIP to the remote sensing domain. The total loss is expressed as

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \mathcal{L}_{\text{ortho}} + \mathcal{L}_{\text{align}}. \tag{9}$$

Given the ground truth referring mask $\mathcal{M} \in \mathbb{R}^{H_2\times W_2}$, the segmentation loss $\mathcal{L}_{\text{seg}}$ is computed between the predicted mask $\hat{\mathcal{M}}$ and $\mathcal{M}$ as

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{ce}}\left(\hat{\mathcal{M}}, \mathcal{M}\right) + \alpha_{\text{dice}}\mathcal{L}_{\text{dice}}\left(\hat{\mathcal{M}}, \mathcal{M}\right). \tag{10}$$

The orthogonal loss $\mathcal{L}_{\text{ortho}}$ is applied to embedding vectors t_1 and t_2 obtained from semantic decomposition [see (4)]. This loss enforces orthogonality by computing the squared cosine similarity between normalized and pooled embeddings to minimize feature dependency

$$\mathcal{L}_{\text{ortho}} = \alpha_{\text{ortho}}\mathcal{L}_{\text{cos-sim}}\left(\Phi_{\text{norm-pool}}\left(t_1\right), \Phi_{\text{norm-pool}}\left(t_2\right)\right) \tag{11}$$

where $\Phi_{\text{norm-pool}}$ represents feature-wise normalization applied to average-pooled vectors, reducing token sequences to single vector representations.

The alignment loss $\mathcal{L}_{\text{align}}$ enhances first-stage CLIP feature extraction capability to improve localization accuracy. This loss incorporates three complementary constraints: dense prompt segmentation ($\mathcal{L}_{\text{dense}}$), spatial-dimension alignment between CLIP visual features v and text semantic features t_{sent} ($\mathcal{L}_{\text{spat}}$), and sample-dimension contrastive learning objective analogous to CLIP ($\mathcal{L}_{\text{samp}}$)

$$\mathcal{L}_{\text{align}} = \alpha_{\text{dense}}\mathcal{L}_{\text{dense}}\left(p_{\text{dense}}, \mathcal{M}\right) + \alpha_{\text{spat}}\mathcal{L}_{\text{spat}}\left(v, t_{\text{sent}}, \mathcal{M}\right) + \alpha_{\text{samp}}\mathcal{L}_{\text{samp}}\left(v, t_{\text{sent}}, \mathcal{M}\right) \tag{12}$$

TABLE I
BENCHMARK DATASETS FOR REFERRING RRSIS

Dataset	Source	Resolution	Size	Triplets	Categories	Attributes
RefSegRS [1]	SkyScapes	0.13 m	512^2	4,420	14	5
RRSIS-D [5]	RSVGD	0.5-30 m	800^2	17,402	20	7
RISBench [3]	VRSBench	0.1-30 m	512^2	52,472	26	8

where $\mathcal{L}_{\text{dense}}$ implements CE constraint on the dense prompt. The spatial alignment loss $\mathcal{L}_{\text{spat}}$ generates coarse masks through Einstein summation of v and t_{sent}, supervised by $\mathcal{M}$ using either hard CE or soft multi-instance learning [20]. The sample alignment loss $\mathcal{L}_{\text{samp}}$ enforces contrastive learning where visual features v undergo annotation-guided mask pooling to form dense representations that constitute positive pairs with t_{sent}. Global contrast is aggregated across computational nodes to ensure training stability [17]. The hyperparameters α control the relative contributions of each component, with their specific effects analyzed in subsequent ablation studies.

IV. Experimental Results and Analyses

A. Experimental Dataset and Settings

The proposed method was evaluated on three referring RRSIS benchmarks: RefSegRS [1], RRSIS-D [5], and RISBench [3]. A statistical comparison of these benchmarks is presented in Table I.

1) RefSegRS [1]: RefSegRS incorporates pixel-level annotations from the SkyScapes dataset [91], integrating diverse referring expressions with corresponding object masks through an automated framework. The dataset comprises 4420 image-description-mask triplets across 285 scenes, partitioned into training (151 scenes, 2172 triplets), validation (31 scenes, 431 triplets), and test (103 scenes, 1817 triplets) subsets. Annotations encompass 14 hierarchical semantic categories, including roads, buildings, cars, and vans, with each object characterized by five attributes. All images maintain uniform 512×512 pixel dimensions with 0.13 m/pixel spatial resolution.

2) RRSIS-D [5]: RRSIS-D utilizes bounding box annotations from the RSVGD dataset [92], processed through SAM to generate instance masks. The dataset encompasses 20 semantic categories, including aircraft, golf courses, highway service areas, baseball fields, and stadiums, enhanced by seven additional attributes for semantic richness. Notable scale variation characterizes the dataset, with targets ranging from minimal image areas to over 400 000 pixels. The dataset contains 17 402 image-description-mask triplets, distributed across training (12 181), validation (1740), and test (3481) subsets. Images are resized to 800×800 pixels with spatial resolutions ranging from 0.5 to 30 m/pixel.

3) RISBench [3]: RISBench integrates remote sensing imagery, referring expressions, and visual localization bounding boxes derived from the VRSBench dataset [93]. The dataset consists of 52 472 image-description-mask triplets, partitioned into training (26 300), validation (10 013), and test (16 158) subsets. All images are standardized to a resolution of 512×512 pixels, with spatial resolutions ranging from 0.1 to 30 m/pixel, thereby encompassing a wide spectrum of geographical scales. Semantic annotations comprise 26 distinct

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TABLE II
PERFORMANCE COMPARISON ACROSS VARIOUS EVALUATION METRICS ON THE REFSEGRS TEST DATASET

Method	Publication	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	cIoU	gIoU
LSTM-CNN [94]	ECCV'16	15.69	10.57	5.17	1.10	0.28	53.83	24.76
RRN [95]	CVPR'18	30.26	23.01	14.87	7.17	0.98	65.06	41.88
ConvLSTM [95]	CVPR'18	31.21	23.39	15.30	7.59	1.10	66.12	43.34
CMSA [96]	CVPR'19	28.07	20.25	12.71	5.61	0.83	64.53	41.47
BRINet [97]	CVPR'20	20.72	14.26	9.87	2.98	1.14	58.22	31.51
LSCM [98]	ECCV'20	31.54	20.41	9.51	5.29	0.84	61.27	35.54
CMPC [99]	CVPR'20	32.36	14.14	6.55	1.76	0.22	55.39	40.63
CMPC+ [100]	TPAMI'21	49.19	28.31	15.31	8.12	2.55	66.53	43.65
CRIS [77]	CVPR'22	35.77	24.11	14.36	6.38	1.21	65.87	43.26
LAVT [40]	CVPR'22	51.84	30.27	17.34	9.52	2.09	71.86	47.40
CARIS [101]	ACM MM'23	45.40	27.19	15.08	8.87	1.98	69.74	42.66
RIS-DMMI [102]	CVPR'23	63.89	44.30	19.81	6.49	1.00	68.58	52.15
CrossVLT [103]	TMM'23	71.16	58.28	34.51	16.35	5.06	77.44	58.84
EVF-SAM [104]	Arxiv'24	35.17	22.34	9.36	2.86	0.39	55.51	36.64
CroBIM [3]	Arxiv'24	75.89	61.42	34.07	12.99	2.75	72.33	59.77
LGCE [1]	TGRS'24	73.75	61.14	39.46	16.02	5.45	76.81	59.96
DANet [14]	ACM MM'24	76.61	64.59	42.72	18.29	8.04	79.53	62.14
RMSIN [5]	CVPR'24	79.20	65.99	42.98	16.51	3.25	75.72	62.58
FIANet [6]	TGRS'24	84.09	77.05	61.86	33.41	7.10	78.32	68.67
SBANet [25]	JPRS'25	77.02	-	44.15	-	8.97	79.86	62.73
BTDNet [12]	Arxiv'25	83.60	75.07	62.69	34.40	9.14	80.57	67.95
DiffRIS [4]	Arxiv'25	74.57	64.56	46.51	24.49	5.78	76.18	62.38
Enti-TwistNet [27]	TGRS'25	77.72	69.30	45.01	26.66	8.49	78.98	63.82
SegEarth-R1 [38]	Arxiv'25	86.30	79.53	69.57	48.87	10.73	79.00	72.45
RS2-SAM 2 [28]	Arxiv'25	84.31	79.42	70.89	55.70	21.19	80.87	73.90
RSRefSeg 2 (Ours)	-	**88.22**	**82.99**	**73.97**	**60.92**	**34.40**	**81.24**	**77.39**

categories, each augmented by eight attributes to support multigranular analysis. The referring expressions exhibit an average length of 14.31 words and a vocabulary of 4431 unique terms, reflecting considerable lexical diversity and syntactic complexity.

B. Evaluation Protocol and Metrics

Following established practices in referring segmentation [1], [5], evaluation is conducted using generalized intersection over union (gIoU) and cumulative intersection over union (cIoU), formulated as

$$\text{gIoU} = \frac{1}{N}\sum_{i=1}^{N}\frac{|P_i \cap G_i|}{|P_i \cup G_i|}$$
$$\text{cIoU} = \frac{\sum_{i=1}^{N}|P_i \cap G_i|}{\sum_{i=1}^{N}|P_i \cup G_i|} \tag{13}$$

where P_i and G_i represent the predicted and ground-truth segmentation masks for the ith image, respectively, and N denotes the total number of images. While both metrics are reported, the primary analysis focuses on gIoU given cIoU's inherent bias toward larger target regions.

Segmentation quality is further assessed using precision at IoU thresholds (Pr@X), which quantifies the proportion of test images achieving IoU values exceeding specified thresholds $X \in \{0.5, 0.6, \ldots, 0.9\}$.

C. Implementation Details

RSRefSeg 2 represents a decoupled paradigm for referring segmentation that leverages the complementary strengths of two pretrained foundation models: CLIP's cross-modal alignment capabilities and SAM's high generalization capacity for segmentation [53], [105]. Unless stated otherwise, the SAM variant employed is "sam2.1-hiera-large,[1]" and the CLIP variant utilized is "siglip2-so400m-patch16-512.[2]"

1) Architecture Details: To address the domain shift between general-purpose vision models and remote sensing applications, low-rank fine-tuning parameters with $r = 16$ are integrated into the backbone networks. The lightweight mask decoder from SAM is also incorporated within the fine-tuning. For referring semantic decomposition, $n_t = 3$ embedding vectors are assigned to distinct semantic subspaces, while sparse-dense prompt generation employs $n_p = 9$ preset sparse prompt embeddings. The preset subspace embedding vectors (t_1^q and t_2^q) are initialized through pooling operations on original referring text representations t_{word}, with learnable positional embeddings subsequently integrated during attention operations. The preset sparse prompt query vectors undergo random initialization and similarly incorporate learnable positional embeddings. Unless specified otherwise, the number of blocks is uniformly set to $N_{\text{decomp}} = N_{\text{interact}} = N_{\text{p-gen}} = 2$ across all modules. Although SAM generates

[1]https://huggingface.co/facebook/sam2.1-hiera-large
[2]https://huggingface.co/google/siglip2-so400m-patch16-512

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TABLE III
PERFORMANCE COMPARISON ACROSS VARIOUS EVALUATION METRICS ON THE RRSIS-D TEST DATASET

Method	Publication	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	cIoU	gIoU
RRN [95]	CVPR'18	51.07	42.11	32.77	21.57	6.37	66.43	45.64
CMSA [96]	CVPR'19	55.32	46.45	37.43	25.39	8.15	69.39	48.54
CMPC [99]	CVPR'20	55.83	47.40	36.94	25.45	9.19	69.22	49.24
LSCM [98]	ECCV'20	56.02	46.25	37.70	25.28	8.27	69.05	49.92
BRINet [97]	CVPR'20	56.90	48.77	39.12	27.03	8.73	69.88	49.65
CMPC+ [100]	TPAMI'21	57.65	47.51	36.97	24.33	7.78	68.64	50.24
LAVT [40]	CVPR'22	69.52	63.63	53.29	41.60	24.94	77.19	61.04
RIS-DMMI [102]	CVPR'23	68.74	60.96	50.33	38.38	21.63	76.20	60.12
CrossVLT [103]	TMM'23	70.38	63.83	52.86	42.11	25.02	76.32	61.00
LGCE [1]	TGRS'24	67.65	61.53	51.45	39.62	23.33	76.34	59.37
EVF-SAM [104]	Arxiv'24	72.16	66.50	56.59	43.92	25.48	76.77	62.75
FIANet [6]	TGRS'24	74.46	66.96	56.31	42.83	24.13	76.91	64.01
RMSIN [5]	CVPR'24	74.26	67.25	55.93	42.55	24.53	77.79	64.20
CroBIM [3]	Arxiv'24	74.58	67.57	55.59	41.63	23.56	75.99	64.46
CADFormer [16]	JSTARS'25	74.20	67.62	55.59	42.37	23.59	77.26	63.77
LSCF [9]	TGRS'25	74.30	67.69	56.32	43.08	25.67	77.42	64.25
RSRefSeg-l [2]	IGARSS'25	74.49	68.33	58.73	48.50	30.80	77.24	64.67
SBANet [25]	JPRS'25	75.91	-	57.05	-	25.38	79.22	65.52
BTDNet [12]	Arxiv'25	75.93	69.92	59.29	46.25	27.46	79.23	66.04
DiffRIS [4]	Arxiv'25	71.79	65.47	54.64	42.89	26.46	76.05	62.19
Enti-TwistNet [27]	TGRS'25	77.94	68.81	57.07	44.63	24.80	78.16	66.83
RSTG-SAM [30]	GRSL'25	76.76	72.11	62.11	47.80	29.73	77.12	65.89
SegEarth-R1 [38]	Arxiv'25	76.96	-	-	-	-	78.01	66.40
RS2-SAM 2 [28]	Arxiv'25	77.56	72.34	61.76	47.92	29.73	78.99	66.72
RSRefSeg 2 (Ours)	-	**80.23**	**75.78**	**65.41**	**50.65**	**31.05**	**79.45**	**69.17**

multiple output masks, only the first is selected as the final prediction.

2) Training Details: Training images are maintained at original dimensions of $H_1 = W_1 = 512$ and $H_2 = W_2 = 1024$, consistent with SAM and CLIP input requirements. Data augmentation is not employed during training. Fine-tuning is applied exclusively to newly introduced low-rank parameters, cascaded prompter parameters, and SAM decoder parameters, while all other network components remain frozen. Loss balancing coefficients are set as $\alpha_{\text{dice}} = 1$, $\alpha_{\text{ortho}} = \alpha_{\text{samp}} = 0.5$, and $\alpha_{\text{dense}} = \alpha_{\text{spat}} = 0$. Optimization is performed using AdamW with an initial learning rate of 1×10^{-4}, batch size of 64, and a maximum of 300 epochs. Learning rate decay follows a Cosine Annealing Scheduler with linear warm-up. All experiments are conducted on NVIDIA H800 GPUs using the OpenMMLab[3] platform and PyTorch framework, with newly introduced modules trained from scratch. Training efficiency is enhanced through BF16 precision under the DeepSpeed ZeRO Stage 2 distributed framework.

D. Comparison With the State-of-the-Art

Comparative evaluation of RSRefSeg 2 is conducted against several state-of-the-art referring image segmentation methods across three categories: general vision approaches (e.g., CMPC [99], LAVT [40], and CrossVLT [103]), remote sensing specific methods (e.g., RMSIN [5], FIANet [6], and SBANet [25]), and foundation model-based approaches (e.g., RSRefSeg [2], EVF-SAM [104], DiffRIS [4], Enti-TwistNet [30], RSTG-SAM [30], RS2-SAM 2 [28], and SegEarth-R1 [38]). The highest values for each metric are highlighted in gray within the tables.

1) Quantitative Results on the RefSegRS Dataset: Table II presents comparative performance analysis on the RefSegRS test dataset. Experimental results demonstrate that RSRefSeg 2 achieves superior performance across all evaluated metrics, with consistently higher precision than existing methods at varying IoU thresholds. While recent contemporary approaches, including RS2-SAM (which also adopts SAM as a decoder), SegEarth-R1, and BTDNet, have significantly advanced referring image segmentation in remote sensing, RSRefSeg 2 comprehensively outperforms these contemporary solutions. Notably, RSRefSeg 2 attains 81.24% cIoU and 77.39% gIoU on composite metrics. The innovative design advances the state of the art, particularly excelling in precise segmentation tasks requiring higher IoU thresholds.

2) Quantitative Results on the RRSIS-D Dataset: The RRSIS-D dataset exceeds RefSegRS in scale, encompassing 20 categories with approximately a fourfold increase in triplets. As shown in Table III, RSRefSeg 2 achieves optimal performance across all evaluation metrics. While performance degradation is observed across methods as thresholds increase, RSRefSeg 2 exhibits comparatively smaller degradation, indicating superior segmentation accuracy and robustness.

[3]https://github.com/open-mmlab/mmsegmentation

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TABLE IV
PERFORMANCE COMPARISON ACROSS CATEGORIES ON THE RRSIS-D TEST DATASET BY CLASS-SPECIFIC AND MEAN IOU

Category	LAVT [40]	RMSIN [5]	LGCE [1]	FIANet [6]	RSRefSeg 2 (Ours)
airport	66.44	68.08	68.11	68.66	**72.95**
golf field	56.53	56.11	56.43	57.07	**79.09**
expressway service area	76.08	76.68	77.19	**77.35**	76.49
baseball field	68.56	66.93	70.93	70.44	**87.27**
stadium	81.77	83.09	84.90	84.87	**88.73**
ground track field	81.84	81.91	**82.54**	82.00	79.88
storage tank	71.33	73.65	73.33	76.99	**79.05**
basketball court	70.71	72.26	74.37	74.86	**74.91**
chimney	65.54	68.42	68.44	68.41	**83.97**
tennis court	74.98	76.68	75.63	78.48	**79.98**
overpass	66.17	**70.14**	67.67	70.01	66.17
train station	57.02	62.67	58.19	61.30	**67.74**
ship	63.47	64.64	63.48	65.96	**71.65**
expressway toll station	63.01	65.71	61.63	64.82	**75.03**
dam	61.61	68.70	64.54	**71.31**	65.83
harbor	60.05	60.40	60.47	**62.03**	46.79
bridge	30.48	36.74	34.24	37.94	**55.56**
vehicle	42.60	47.63	43.12	49.66	**55.10**
windmill	35.32	41.99	40.76	46.72	**62.06**
airplane	55.29	60.17	56.43	60.32	**72.95**
average	62.44	65.13	64.12	66.46	**72.06**

TABLE V
PERFORMANCE COMPARISON ACROSS VARIOUS EVALUATION METRICS ON THE RISBENCH TEST DATASET

Method	Publication	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	cIoU	gIoU
RRN [95]	CVPR'18	55.04	47.31	39.86	32.58	13.24	49.67	43.18
MAttNet [106]	CVPR'18	56.83	48.02	41.75	34.18	15.26	51.24	45.71
BRINet [97]	CVPR'20	52.87	45.39	38.64	30.79	11.86	48.73	42.91
LSCM [98]	ECCV'20	55.26	47.14	40.10	33.29	13.91	50.08	43.69
CMPC [99]	CVPR'20	55.17	47.84	40.28	32.87	14.55	50.24	43.82
CMPC+ [100]	TPAMI'21	58.02	49.00	42.53	35.26	17.88	53.98	46.73
CRIS [77]	CVPR'22	63.67	55.73	44.42	28.80	13.27	69.11	55.18
LAVT [40]	CVPR'22	69.40	63.66	56.10	44.95	25.21	74.15	61.93
ETRIS [107]	ICCV'23	60.98	51.88	39.87	24.49	11.18	67.61	53.06
CrossVLT [103]	TMM'23	70.62	65.05	57.40	45.80	26.10	74.33	62.84
RIS-DMMI [102]	ICCV'23	72.05	66.48	59.07	47.16	26.57	74.82	63.93
CARIS [101]	ACM MM'23	73.94	68.93	62.08	50.31	29.08	75.10	65.79
robust-ref-seg [108]	TIP'24	69.15	63.24	55.33	43.27	24.20	74.23	61.25
LGCE [1]	TGRS'24	69.64	64.07	56.26	44.92	25.74	73.87	62.13
RMSIN [5]	CVPR'24	71.01	65.46	57.69	45.50	25.92	74.09	63.07
CroBIM-Swin [3]	Arxiv'24	75.75	70.34	63.12	51.12	28.45	73.61	67.32
CroBIM-ConvNeXt [3]	Arxiv'24	77.55	72.83	66.38	55.93	34.07	73.04	69.33
DiffRIS [4]	Arxiv'25	74.50	69.42	62.96	53.26	34.57	73.00	67.04
Enti-TwistNet [27]	TGRS'25	76.13	71.47	63.48	52.38	27.46	74.07	67.50
LSCF [9]	TGRS'25	76.08	71.29	64.96	55.13	36.73	**74.88**	68.53
RSRefSeg 2 (Ours)	-	**79.08**	**75.36**	**70.62**	**62.75**	**48.75**	74.77	**72.57**

Table IV displays performance across different categories, where RSRefSeg 2 achieves 72.06% mean IoU, surpassing FIANet (66.46%). Optimal results are yielded in the majority (15 of 20) categories, with particularly strong performance on targets characterized by complex structures yet distinct boundaries (e.g., golf courses, baseball fields, stadiums, and chimneys). Substantial improvements over alternative methods are demonstrated on challenging small object categories such as bridges, vehicles, and windmills, indicating exceptional fine-grained detail capture capability. Minor deficiencies are observed for categories such as dam and harbor, likely attributed to ambiguous boundaries that amplify segmentation difficulty.

3) Quantitative Results on the RISBench Dataset: RISBench extends the dataset scale by incorporating 26 common ground object categories with broader image resolutions and

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Fig. 4. Segmentation examples of the proposed RSRefSeg 2 on the RefSegRS test dataset.

eight attribute categories, posing greater referring segmentation challenges. As detailed in Table V, RSRefSeg 2 achieves optimal performance across nearly all evaluation metrics, demonstrating significant improvement over existing state-of-the-art methods. The method comprehensively surpasses comparative approaches in precision across different threshold intervals, exhibiting distinct advantages under higher threshold conditions for enhanced fine-grained segmentation accuracy. While achieving comparable performance to the recently proposed LSCF method on cIoU, RSRefSeg 2 substantially outperforms it on gIoU.

4) Qualitative Visualizations: Representative segmentation instances from each dataset are illustrated in Figs. 4–6, demonstrating correspondence between input images, textual descriptions, ground truth masks, and model predictions. The model effectively interprets complex referring instructions, including spatial descriptions (e.g., "at the top right") and color attributes to achieve precise target segmentation. Accurate segmentation is demonstrated across diverse object categories, including sports venues (e.g., track fields and baseball fields), infrastructure (e.g., storage tanks, bridges, and airports), and vehicles (e.g., cars and buses). High-quality segmentation masks are produced with strong alignment to ground truth annotations, characterized by clear edge contours and complete coverage. Multiscale processing capability enables segmentation of both large-scale targets (e.g., track fields and airports) and fine-grained small targets (e.g., vehicles). Excellent detail discrimination ability is demonstrated through fine-grained perception and semantic understanding, particularly for tasks requiring distinction between similar objects (e.g., "the aircraft to the left of the gray plane" or "the ship slightly larger than the ship on the left").

E. Ablation Study

Ablation studies are conducted on the RRSIS-D dataset to systematically evaluate the contributions of individual components and parameter configurations within the proposed framework. Unless otherwise specified, referring descriptions are encoded by SigLip [105] as semantic prompts, while referring mask extraction is performed by SAM2.1-s [53].

1) Effects of Tuning Parameters in RSRefSeg 2 Encoders: RSRefSeg 2 leverages CLIP's multimodal understanding to achieve coarse localization of textual descriptions and generate semantic prompts, while employing SAM's segmentation capability to refine precise masks. Although CLIP and SAM,

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Fig. 5. Segmentation examples of the proposed RSRefSeg 2 on the RRSIS-D test dataset.

as billion-parameter vision foundation models, demonstrate exceptional generalization, directly training all parameters risks severe overfitting, while fully freezing them exacerbates domain shift effects. To address this challenge, a LoRA strategy is adopted, selectively modifying parameters within the Transformer layers of each encoder. The impact of varying LoRA ranks on the CLIP text encoder (clip-t), CLIP vision encoder (clip-v), and SAM vision encoder (sam-v) is analyzed in Table VI. A rank of 0 indicates fully frozen parameters, while the optimal configuration (†) employs ranks of 16, 16, and 32 for clip-t, clip-v, and sam-v, respectively. For each encoder, only its rank is varied while others remain fixed at optimal values.

Several key observations emerge as follows.

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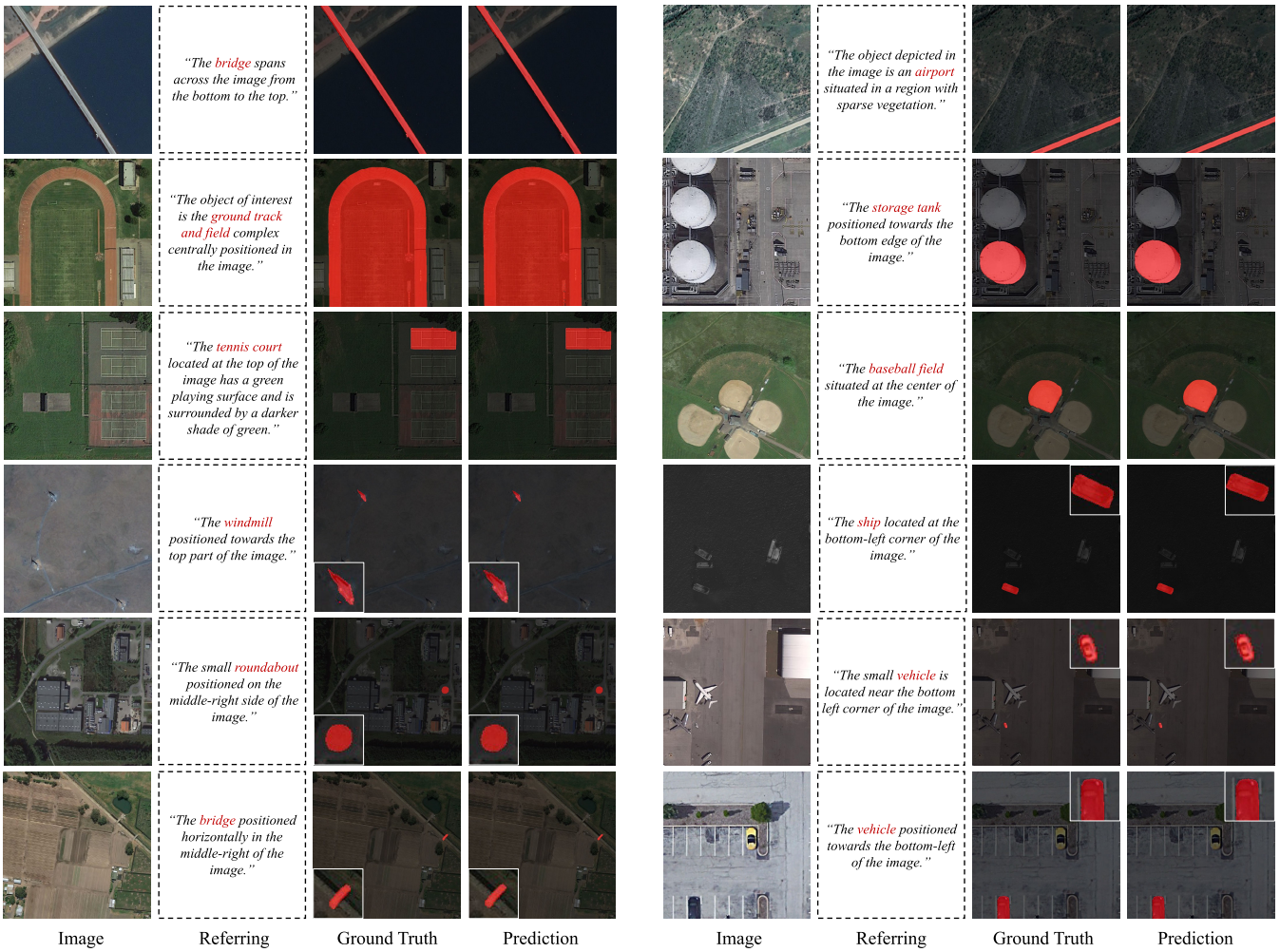


Fig. 6. Segmentation examples of the proposed RSRefSeg 2 on the RISBench test dataset.

1) Parameter fine-tuning significantly enhances performance, with vision encoders (clip-v and sam-v) yielding greater gains than the text encoder (clip-t). This occurs because vision encoders, pretrained on natural images, experience a substantial domain shift in remote sensing contexts, whereas linguistic descriptions exhibit minimal divergence across domains.
2) Balancing trainable parameters is critical; insufficient adaptation impedes cross-domain transfer, while excessive parameters cause overfitting despite faster convergence.
3) Vision encoders require higher ranks than text encoders, reflecting greater visual domain shift and the increased complexity of remote sensing imagery.
4) Performance exhibits higher sensitivity to rank variations in the CLIP vision encoder than in SAM, underscoring the importance of semantic prompt precision, as SAM maintains robust segmentation generalizability under limited parameter adjustments. The SAM visual encoder employs a rank of 32 for SAM2.1-s and 16 for SAM2.1-l. This approach trains only ~98-M parameters (LoRA components + cascaded prompter + SAM decoder) from the total 1.4B parameters, achieving approximately 7% parameter efficiency.

2) Effects of Varying Module Depths on the Cascaded Referring Prompter: The prompter comprises three sequential modules: referring semantic decomposition, text–visual interaction, and sparse-dense prompt generation. Each module incorporates serially connected blocks, with block numbers denoted as N_{decomp}, N_{interact}, and $N_{\text{p-gen}}$, respectively (see Section III-C). The impact of varying block counts is presented in Table VII, where optimal performance is marked with †. Experimental results demonstrate that employing two blocks for all three modules achieves an optimal trade-off between model complexity and performance. Insufficient blocks constrain representational capability, while excessive blocks increase computational overhead and overfitting risk.

3) Effects of Embedding Vector Count in Semantic Decomposition and Prompt Generation: Query embeddings are introduced to represent semantic subspaces during decomposition and sparse prompts during prompt generation, aggregating textual semantic features and prompt positional information, respectively. The model's expressive capacity correlates positively with embedding quantity, denoted as n_t for textual queries and n_p for prompt queries (see Section III-C).

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TABLE VI

PERFORMANCE WITH VARYING LORA RANKS FOR CLIP-T, CLIP-V, AND SAM-V ENCODERS

Enc.	Rank	Pr@0.5	Pr@0.7	Pr@0.9	cIoU	gIoU
clip-t	0	72.16	55.93	27.92	77.80	63.40
	4	73.48	57.94	28.69	78.65	64.61
	8	74.20	57.79	28.41	78.44	64.82
	16†	76.60	60.25	28.58	78.15	66.01
	32	73.63	57.94	29.50	79.01	64.53
	64	72.76	57.16	28.35	78.43	64.14
clip-v	0	65.48	50.90	26.68	75.96	57.86
	4	73.42	58.20	29.81	78.44	64.64
	8	73.83	57.22	29.18	78.64	64.33
	16†	76.60	60.25	28.58	78.15	66.01
	32	75.23	58.46	29.07	79.36	65.39
	64	73.97	57.85	29.30	78.96	64.88
sam-v	0	69.52	52.31	24.82	76.91	60.82
	4	73.57	57.88	29.15	78.60	64.64
	8	73.71	57.82	28.78	78.38	64.67
	16	74.46	58.23	28.87	78.62	65.05
	32†	76.60	60.25	28.58	78.15	66.01
	64	75.00	58.23	29.27	78.98	65.44

TABLE VII

PERFORMANCE OF VARYING BLOCK COUNTS ACROSS MODULES IN THE CASCADED SECOND-ORDER REFERRING PROMPTER

Module	Blocks	Pr@0.5	Pr@0.7	Pr@0.9	cIoU	gIoU
Decomp.	1	73.77	58.11	29.07	78.36	64.72
	2†	76.60	60.25	28.58	78.15	66.01
	4	74.43	58.46	29.47	78.91	65.20
	6	74.46	58.32	28.81	78.73	65.14
Interac.	1	74.46	58.97	29.10	78.51	64.86
	2†	76.60	60.25	28.58	78.15	66.01
	4	74.08	57.71	28.95	78.77	64.65
	6	73.31	57.42	28.52	78.06	64.29
Prompt	1	74.40	58.51	29.18	78.95	65.04
	2†	76.60	60.25	28.58	78.15	66.01
	4	74.08	58.03	29.44	78.43	65.00
	6	74.60	58.54	29.13	78.49	64.88

TABLE VIII

PERFORMANCE OF VARYING NUMBERS OF EMBEDDING VECTORS IN SEMANTIC DECOMPOSITION AND PROMPT GENERATION. OPTIMAL CONFIGURATIONS ARE MARKED WITH †

Module	Embeddings	Pr@0.5	Pr@0.7	Pr@0.9	cIoU	gIoU
Decomp.	1	74.66	58.46	28.87	78.28	65.26
	3†	76.60	60.25	28.58	78.15	66.01
	6	75.15	58.48	29.38	78.96	65.18
	9	74.66	56.88	29.38	77.96	64.09
Prompt	1	73.59	57.59	29.21	78.66	64.49
	3	74.37	57.65	28.21	78.30	64.51
	6	73.71	57.88	29.10	78.33	64.72
	9†	76.60	60.25	28.58	78.15	66.01
	12	74.46	57.82	29.10	79.11	65.22

As demonstrated in Table VIII, optimal performance for the decomposition module is achieved with 3 embeddings representing distinct semantic subspaces, while the prompt generation module performs optimally with nine embeddings for sparse prompt representation. Although optimal settings are dataset-dependent, with empirical evidence indicating that increasing n_t and n_p enhances expressiveness for complex

TABLE IX

PERFORMANCE OF POSITIONAL ENCODING AND EMBEDDING METHODS ACROSS PROMPTER MODULES

Module	PE	Emb.	Pr@0.5	Pr@0.7	Pr@0.9	cIoU	gIoU
Decomp.	✗	learnable	73.02	57.25	28.00	77.72	64.09
	✗	textpool	74.03	58.26	28.55	78.27	64.82
	✓	zero	74.86	58.28	29.38	79.34	65.57
	✓	learnable	74.08	58.08	28.98	78.27	64.86
	✓	textpool	76.60	60.25	28.58	78.15	66.01
Interac.	✗	-	74.57	58.11	28.46	77.94	64.80
	✓	sine	74.26	58.23	28.98	77.90	64.66
	✓	learnable	76.60	60.25	28.58	78.15	66.01
Prompt	✗	learnable	74.54	58.14	29.13	79.11	64.98
	✓	zero	75.15	57.68	28.98	78.85	65.12
	✓	learnable	76.60	60.25	28.58	78.15	66.01

TABLE X

PERFORMANCE ON DIFFERENT PROMPTER ARCHITECTURES

Architecture	Pr@0.5	Pr@0.7	Pr@0.9	cIoU	gIoU
Baseline (Single-stage Mask Prompt)	64.63	45.49	16.93	59.62	57.39
Baseline (Single-stage Prompter)	67.88	52.57	25.39	74.11	60.06
Parallel Interaction	70.64	54.38	24.99	72.75	61.60
Cascaded (Ours)	72.99	57.39	28.12	79.22	64.13
Cascaded (w/ Dense Prompt, Ours)	76.60	60.25	28.58	78.15	66.01

textual descriptions or high-instance images, while reduction suffices for simpler cases, unified configurations of $n_t = 3$ and $n_p = 9$ are adopted without exhaustive per-dataset tuning. This approach achieves exceptional performance across all three datasets, outperforming existing comparative methods and demonstrating the proposed architecture's robustness across diverse application scenarios.

4) Effects of Various Component Configurations in the Prompter: This section analyzes the effects of incorporating positional encoding and various embedding methods. The notation employs "✔" and "✘" to indicate inclusion and absence of positional encoding, respectively, while embedding methods include "zero" (zero-initialized vectors), "learnable" (randomly sampled learnable vectors), "textpool" (pooled original text features), and "sine" (fixed sine/cosine coordinate-based vectors).

Table IX reveals that positional encoding is critical across all modules, with its inclusion ("✔") consistently yielding significant performance improvements. Substantial gains are observed when positional encoding is added to the decomposition module, regardless of the embedding method employed. For the referring semantic decomposition module, optimal performance is achieved using "textpool" embeddings combined with positional encoding, followed by "zero" initialized embeddings with positional encoding, while learnable vectors prove suboptimal. Within the text–visual interaction module, "learnable" embeddings demonstrate optimal effectiveness when positional encoding is applied to visual feature maps, whereas "sine" positional encoding shows no significant advantage. The prompt generation module achieves optimal performance through 'learnable' embeddings combined with positional encoding.

5) Impact of Prompter Architecture: To investigate the rationale behind the cascaded second-order referring prompter

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TABLE XI
PERFORMANCE OF INTRASUBSPACE SELF-ATTENTION AND INTERSUBSPACE CROSS-ATTENTION

intra-attn	inter-attn	Pr@0.5	Pr@0.7	Pr@0.9	cIoU	gIoU
✗	✗	72.68	57.74	29.08	78.16	64.08
✓	✗	74.23	58.20	29.09	78.27	64.86
✓	✓	76.60	60.25	28.58	78.15	66.01

TABLE XII
PERFORMANCE OF DIFFERENT ATTENTION MECHANISM PAIRINGS IN THE TEXT–VISUAL INTERACTION

self-attn	cross-attn	Pr@0.5	Pr@0.7	Pr@0.9	cIoU	gIoU
✗	✗	73.54	56.27	28.08	77.82	64.39
✓	✗	73.22	57.42	28.28	78.03	64.54
✓	✓	76.60	60.25	28.58	78.15	66.01

TABLE XIII
PERFORMANCE OF VARYING DENSE PROMPT CONSTRUCTION METHODS

Setting	Pr@0.5	Pr@0.7	Pr@0.9	cIoU	gIoU
T-t_2	73.80	58.05	28.69	77.79	64.44
T-t_2'	71.07	55.55	28.21	76.33	62.58
T-sparse	71.10	56.42	27.92	76.60	63.08
T-t_{sent}	76.60	60.25	28.58	78.15	66.01
w/ T-soft	74.51	57.65	28.84	77.86	64.74
w/o T-soft	76.60	60.25	28.58	78.15	66.01
V-v	73.80	57.82	28.92	78.67	64.93
V-v_2'	76.60	60.25	28.58	78.15	66.01
w/o V-soft	74.11	59.20	29.55	78.58	65.02
w/ V-soft	76.60	60.25	28.58	78.15	66.01
post-up	74.66	59.83	28.89	77.47	65.21
pre-up	76.60	60.25	28.58	78.15	66.01

and validate our core architectural design choices, we compare different architectural variants in Table X.

a) Baseline (Single-Stage Mask Prompt): This configuration removes the learnable prompter network. Instead, the coarse segmentation mask derived directly from CLIP text embeddings (t_{sent}) and visual features (v) serves as the mask prompt for SAM fine-tuning. The inferior performance indicates that raw semantic masks from CLIP are insufficient to effectively guide SAM.

b) Baseline (Single-Stage Prompter): This variant removes the semantic decomposition and cascaded interaction. The holistic CLIP text embedding interacts directly with visual features in a single stage. To ensure a fair comparison, we scale up the attention layers to match the parameter budget of the cascaded prompter. The suboptimal gIoU (60.06%) confirms that a holistic embedding struggles to distinguish targets in multientity scenarios due to semantic ambiguity.

c) Parallel Interaction: Text embeddings are decomposed into two subspaces that interact with visual features independently (in parallel) before being fused via element-wise addition. Although this method slightly outperforms the baselines, it falls short of the cascaded design. This suggests that without the spatial prior established by the first stage, the attribute and spatial features in the second subspace struggle to accurately localize the target within the global context.

d) Cascaded (Ours): Our proposed sequential design achieves superior performance. The cascaded structure significantly improves results, verifying our hypothesis that the first stage effectively anchors potential targets, enabling the second stage to focus on refining the selection based on attributes and spatial relations, thereby progressively eliminating misactivated regions. Furthermore, integrating dense prompts into the cascaded architecture yields the highest performance (66.01% gIoU), demonstrating that dense cues effectively augment sparse prompt generation.

It is worth noting that the "Cascaded" performance relies on the semantic independence of the subspaces. As detailed in the loss function analysis (see Section IV-E10), removing the orthogonal constraint ($\mathcal{L}_{\text{ortho}}$) from the cascaded structure results in a performance drop (from 66.01% to approximately 64.21%), which confirms the necessity of this constraint for the cascaded architecture to function effectively.

6) Effects of Subspace Attention Mechanism in Referring Semantic Decomposition: The semantic decomposition approach facilitates implicit cascaded reasoning through three principal mechanisms operating on dual semantic subspaces derived from the original referring text: intrasubspace self-attention within each semantic subspace, cross-attention bridging subspaces with original text embeddings, and external cross-attention enabling intersubspace interaction [see (4)]. As shown in Table XI, optimal performance is achieved when intrasubspace self-attention is combined with intersubspace cross-attention (final row), underscoring the necessity of bidirectional information exchange for mutual subspace comprehension and effective semantic fusion. The intrasubspace self-attention mechanism proves essential for establishing accurate semantic representations within individual subspaces and enabling subspace-specific modeling capabilities.

7) Effects of Attention Mechanism in Text–Visual Interaction: Text–visual interaction employs two core attention mechanisms: self-attention for processing features within individual semantic subspaces, and cross-attention for facilitating interaction between subspaces and semantically enhanced visual features. As demonstrated in Table XII, cross-attention yields the most significant performance improvements, demonstrating its effectiveness in enhancing vision-language integration and strengthening multimodal feature representation. Conversely, self-attention alone produces only marginal accuracy gains.

8) Effects of Dense Prompt Construction Methods: Dense and sparse prompts jointly provide target location information for the second-stage fine-grained segmentation. As formalized in (7), dense prompts utilize text embedding vectors as channel-wise filters to weight visual features. The effectiveness of dense prompts, verified in Table X, motivated this comprehensive evaluation of their compositional variants, as summarized in Table XIII.

i) Four text embedding sources were evaluated: T-t_2 from the pooled second semantic subspace vector t_2 [see (4)]; T-t_2' from the pooled vector t_2' after visual-text interaction [see (5)]; T-sparse from the additional token filter in sparse

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TABLE XIV
PERFORMANCE OF VARYING INPUT RESOLUTIONS OF CLIP VISUAL ENCODER

Size	Pr@0.5	Pr@0.7	Pr@0.9	cIoU	gIoU
224^2	76.47	60.21	28.52	78.06	65.85
384^2	76.60	60.25	28.58	78.15	66.01
448^2	76.64	60.07	28.69	78.57	66.10
768^2	76.68	60.29	29.15	78.76	66.14

prompts [see (6)]; and T-t_{sent} as the original global semantic vector t_{sent} (see Section III-B). The original global semantic vector (t_{sent}) achieved superior performance, validating the importance of shallow global text semantics for effective dense prompt generation.

ii) The incorporation of an MLP layer into text embeddings was assessed. Performance degradation observed with MLP processing (w/ T-soft) indicates that additional transformations may disrupt original semantic representations.

iii) Two visual feature sources were compared: V-v corresponding to original CLIP features v (see Section III-B) and V-v_2' representing features v_2' after text-guided interaction [see (5)]. Superior performance was achieved with V-v_2', demonstrating that visual-text interaction effectively refines referring features for improved segmentation.

iv) The application of a 1×1 convolutional layer to visual features was examined. Moderate performance improvement with visual feature processing (w/V-soft) suggests that adaptive soft adjustment enhances visual representations, contrasting with text embedding findings.

v) Given that CLIP feature dimensions ($h_1 < (H_2)/4$, $w_1 < (W_2)/4$) are insufficient for SAM's dense prompt requirements, upsampling strategies were explored. Prefiltering interpolation (Preup) outperformed postfiltering interpolation (Postup), indicating that early stage upsampling better preserves fine-grained information.

The optimal dense prompt configuration employs the original global text semantic vector (t_{sent}) without MLP processing, postinteraction visual features (v_2') with convolutional processing, and prefiltering upsampling. This design harmonizes textual and visual fusion mechanisms to provide accurate spatial guidance for fine-grained segmentation.

9) Effects of Input Size on CLIP Visual Encoder: Table XIV presents model performance across varying input sizes. Results demonstrate consistent improvement in segmentation accuracy with increased input resolution, though the gains remain modest, supporting the hypothesis that coarse localization tasks can be effectively performed at lower resolutions. Considering the computational-performance tradeoff, an input size of 384×384 was selected for the small model variant.

10) Analysis of Loss Function Components: As detailed in Section III-E, the overall loss function integrates several key components. Systematic ablation studies (see Table XV) reveal their distinct contributions to model performance. Under CE segmentation supervision $\mathcal{L}_{\text{ce}}$, introducing the semantic subspace constraint $\mathcal{L}_{\text{ortho}}$ yields substantial metric enhancement, demonstrating its efficacy for segmentation accuracy. However, subsequent integration of dense prompt supervision $\mathcal{L}_{\text{dense}}$

TABLE XV
EFFECTS OF DIFFERENT LOSS FUNCTION COMPONENTS

$\mathcal{L}_{\text{ce}}$	$\mathcal{L}_{\text{ortho}}$	$\mathcal{L}_{\text{dense}}$	$\mathcal{L}_{\text{spatial}}$	$\mathcal{L}_{\text{sample}}$	$\mathcal{L}_{\text{dice}}$	Pr@0.5	Pr@0.7	Pr@0.9	cIoU	gIoU
✓	-	-	-	-	-	79.72	55.41	28.03	78.32	63.16
✓	✓	-	-	-	-	74.62	57.28	28.58	78.37	64.21
✓	✓	✓	-	-	-	73.13	57.82	29.21	78.91	64.08
✓	✓	-	CE-t_{sent}-v	-	-	72.47	58.11	28.98	77.03	63.12
✓	✓	-	MIL-t_{sent}-v	-	-	71.21	57.36	28.98	78.27	63.01
✓	✓	-	CE-t_2-v	-	-	72.56	58.02	29.30	77.54	63.65
✓	✓	-	MIL-t_2-v	-	-	73.54	58.26	28.44	78.80	64.25
✓	✓	-	CE-t_{sent}-v_2'	-	-	72.65	57.51	29.50	76.88	63.77
✓	✓	-	MIL-t_{sent}-v_2'	-	-	73.71	57.65	28.95	77.93	64.02
✓	✓	-	CE-t_2-v_2'	-	-	72.96	58.31	29.53	77.83	63.95
✓	✓	-	MIL-t_2-v_2'	-	-	74.11	59.00	29.53	78.52	64.65
✓	✓	-	-	NCE-t_{sent}-v	-	74.40	58.69	29.24	78.78	65.18
✓	✓	-	-	NCE-t_2-v	-	76.60	60.25	28.58	78.15	66.01
✓	✓	-	-	NCE-t_{sent}-v_2'	-	71.90	56.62	28.46	77.32	63.13
✓	✓	-	-	NCE-t_2-v_2'	-	72.65	57.25	28.09	77.16	63.52
✓	✓	-	-	NCE-t_2-v	✓	77.90	61.13	29.35	78.98	67.06

TABLE XVI
PERFORMANCE COMPARISON OF VARIOUS CLIP AND SAM MODEL COMBINATIONS

CLIP	SAM	Pr@0.5	Pr@0.7	Pr@0.9	cIoU	gIoU
siglip-p14-384	sam2.1-s	76.70	60.35	28.58	78.05	66.13
siglip2-p14-224	sam2.1-s	75.58	59.32	28.26	77.55	65.36
siglip2-p14-224	sam2.1-s	75.58	59.32	28.26	77.55	65.36
siglip2-p14-384	sam2.1-s	77.90	61.13	29.35	78.98	67.06
siglip2-p16-384	sam2.1-s	77.33	60.72	28.66	78.29	66.73
siglip2-p16-512	sam2.1-s	78.94	61.79	29.07	78.43	67.79
siglip2-giant-p16-256	sam2.1-s	76.29	60.27	28.32	77.17	65.62
siglip2-giant-p16-384	sam2.1-s	77.67	60.87	28.82	77.52	66.94
siglip2-p16-512	sam2.1-t	78.56	61.56	27.83	78.48	67.63
siglip2-p16-512	sam2.1-s	78.94	61.79	29.07	78.43	67.79
siglip2-p16-512	sam2.1-b	79.28	63.91	30.45	78.52	68.07
siglip2-p16-512	sam2.1-l	80.23	65.41	31.05	79.45	69.17

produces negligible gains, likely due to direct supervision on low-resolution coarse masks where fine detail loss restricts improvement potential.

Multiple implementations of spatial constraint $\mathcal{L}_{\text{spatial}}$ (e.g., CE-t_{sent}-v, MIL-t_{sent}-v) were evaluated. Both CE and multi-instance learning (MIL) approaches yielded marginal improvements, suggesting the limited efficacy of constraints applied to spatial dimensions of visual feature maps. Conversely, sample-wise NCE learning applied to original CLIP visual features generated substantial performance gains, with NCE-t_2-v configuration demonstrating notable superiority. Finally, integration of segmentation Dice loss $\mathcal{L}_{\text{dice}}$ further enhances model performance, confirming the complementary benefits of region-based optimization for segmentation tasks.

11) Effects of Different Backbone Versions: Table XVI presents various combinations of CLIP and SAM versions and their corresponding performance impacts. Under a fixed SAM configuration (sam2.1-s), increasing the input resolution of CLIP was observed to yield significantly greater performance improvements compared to augmenting its parameter count. This suggests that higher resolution enhances target localization accuracy, which is particularly advantageous in remote sensing scenarios where targets are often small and widely distributed. Conversely, when the CLIP was held constant (siglip2-p16-512), segmentation performance progressively improved with increased SAM parameter size, achieving optimal results using sam2.1-l. These findings confirm the pivotal role of foundation model capacity in the second segmentation stage for enhancing accuracy. Consequently, variants with different specifications may be selected based on computational

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constraints to optimize the performance-to-resource ratio in practical applications.

F. Discussions

Qualitative evaluations (see Figs. 4–6) demonstrate that our model accurately resolves complex spatial instructions (e.g., "upper right" and "lower left") and multiattribute descriptions. The model exhibits superior performance in multiscale object segmentation, diverse category recognition, and boundary delineation, showing particular proficiency with small targets such as bridges and vehicles. This proficiency stems from the dual-gain mechanism of our decoupled architecture: i) during localization, orthogonal subspace projection within the text embedding space effectively mitigates target misactivation caused by multientity descriptions; ii) during segmentation, task decoupling isolates mask generation from text modality interference, ensuring masks maintain sharp edges and structural integrity.

Unlike end-to-end approaches that conflate multimodal alignment with pixel decoding, RSRefSeg 2 enables explicit error attribution. To demonstrate this diagnostic capability, we categorized failure cases on the RRSIS-D dataset into two distinct modes: localization error (IoU $<$ 0.1, indicating target miss) and segmentation error (0.1 $\leq$ IoU $<$ 0.5, indicating accurate localization but imprecise delineation). Quantitative analysis reveals that for our method, localization errors account for 55.5% of failures, while segmentation errors constitute 44.5%. In stark contrast, a baseline single-stage prompter exhibits a distribution of 67.7% (localization) and 32.3% (segmentation). This significant shift confirms that while our cascaded prompter successfully resolves most semantic queries, it thereby reduces localization failures. This insight provides a clear roadmap for future optimization: improving language reasoning for the remaining localization errors and enhancing the mask decoder's edge sensitivity for segmentation errors.

We further present a stage-wise diagnostic analysis (visualized in Fig. 7). Case 1 (localization failure): in scenarios involving compound semantics (e.g., distinguishing "the vehicle on the right" among similar objects), the stage-1 dense prompt may show scattered activation, leading to incorrect masking. We stratified performance by sentence complexity (attribute count extracted via spaCy). Results indicate that our model is highly robust, suffering only a 7.6% accuracy drop on complex queries (>3 attributes) compared to simple ones. Conversely, the single-stage baseline suffers a precipitous 32.1% decline, validating that our orthogonal subspace decomposition effectively mitigates ambiguity. Future work could further reduce residual errors by incorporating stronger reasoning priors, such as a chain-of-thought. Case 2 (segmentation failure): in the third row, while the stage-1 prompt accurately locates the "airport," the stage-2 mask misses fine boundaries. This reflects the limitation of SAM in low-contrast boundary regions (often found at water-land interfaces, see Table IV). Crucially, our decoupled design successfully isolates this as a visual representation issue rather than a semantic failure, contradicting the "black box" nature of coupled methods. Future iterations could address this visual domain gap by exploring domain-specific edge-detection priors or specialized adapter layers without compromising generalization.

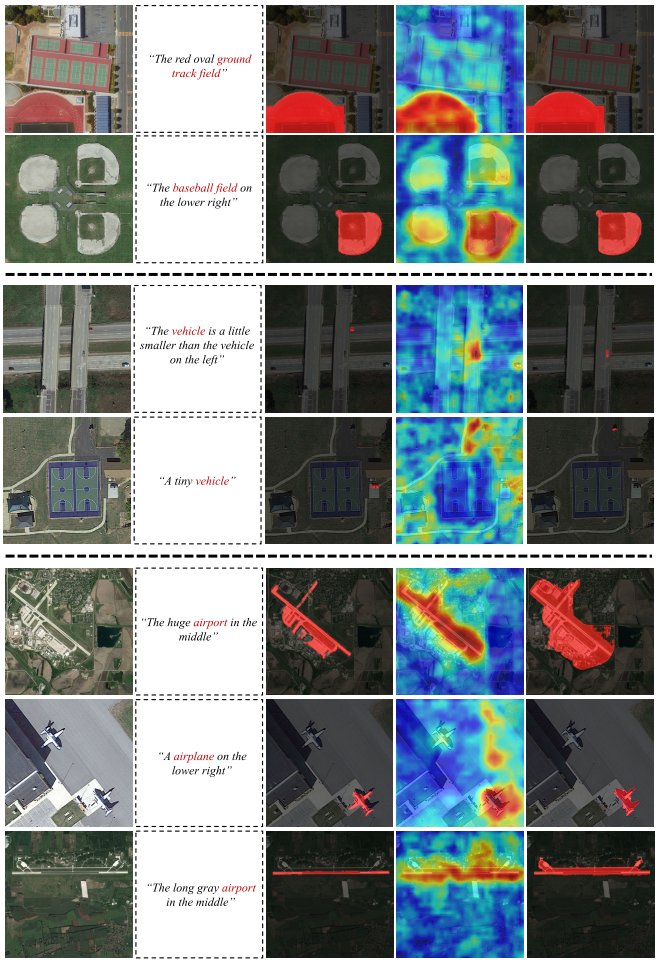


Fig. 7. Segmentation results across various cases. From left to right: input image, text instruction, ground truth, stage-1 output (dense prompt heatmap), and stage-2 output (final prediction).

V. Conclusion

To address the challenges of multimodal feature misalignment, insufficient small target recognition accuracy, and limited model interpretability in referring RRSIS, this article proposes RSRefSeg 2. The method incorporates prior knowledge from visual foundation models through a decoupled two-stage segmentation process comprising coarse localization followed by refined segmentation. A cascaded second-order prompting mechanism is employed to activate text-guided visual features within the pretrained multimodal encoder CLIP, which subsequently directs the foundation segmentation model SAM to generate corresponding masks. In the coarse localization stage, an orthogonality-constrained projection of text embeddings is utilized to mitigate target misactivation caused by conflicting multientity descriptions, thereby enhancing spatial localization precision. The refined segmentation stage implements task decoupling to ensure mask generation proceeds independently of textual comprehension, producing masks with well-defined boundaries and structural integrity. PEFT is achieved through LoRA integration, enabling effective

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adaptation to domain-specific remote sensing knowledge while preserving the intrinsic general representation capabilities of pretrained models using only 7% of the total training parameters. Experimental validation on RefSegRS, RRSIS-D, and RISBench demonstrates that RSRefSeg 2 achieves significant performance improvements, particularly in open-scene semantic understanding, multiscale object segmentation, small target recognition (e.g., ships and vehicles), and complex spatial relationship parsing. By explicitly distinguishing between localization and segmentation errors, RSRefSeg 2 overcomes the error propagation and interpretability limitations inherent in traditional fusion-then-decoding frameworks, providing reliable support for advancing remote sensing intelligent interpretation paradigms.

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